A History-Conditioned Cross-Modal Attention Framework for Multimodal Lung Disease Diagnosis from Chest X-Rays

Arnav Panwar 23BAI0147, David Kumar 23BAI0199
VIT University, Vellore

***Abstract*—This work aims to improve chest X-ray–based lung disease diagnosis by enabling patient-specific reasoning through the joint modeling of imaging evidence and free-text clinical history. The proposed system is designed as a clinical decision-support framework for multi-label thoracic disease diagnosis using chest X-rays and associated patient medical histories, with a particular focus on large-scale datasets such as NIH ChestX-ray14. We propose a history-conditioned multimodal diagnostic architecture that integrates a lung-focused visual encoder with a ClinicalBERT-based clinical history encoder using token-level cross-modal attention and a learned consistency mechanism, allowing clinical history to dynamically refine visual interpretation rather than serving as an independent decision source. The core contribution is an early fusion strategy that performs token-level image–history alignment to learn history-aware visual attention and diagnostic consistency, replacing conventional decision-level agreement between separate predictors. Compared to existing image-only models, structured clinical fusion methods, and late-fusion multimodal approaches, the proposed technique enables richer patient-specific reasoning, reduces visually plausible but clinically inconsistent predictions, improves interpretability through explicit image–history interactions, and avoids reliance on paired radiology reports or manually engineered clinical features. As of now, experimental results are not available due to which performance evaluation cannot be done for this.**

Keywords—Multimodal medical diagnosis, Chest X-ray analysis, Clinical history modeling, Vision language fusion, AutoEncoders, Cross modal attention.

I. Introduction

Chest X-ray imaging is one of the most widely used diagnostic tools for the early detection and screening of thoracic diseases due to its low cost, speed, and accessibility. Automated chest X-ray analysis systems have therefore gained significant attention as clinical decision-support tools to assist radiologists in identifying lung abnormalities, particularly in high-volume and resource-constrained healthcare settings.

Despite recent advances, most existing approaches either rely solely on imaging data or incorporate clinical information through structured variables or late decision-level fusion, where predictions from image and text models are combined post hoc. Such strategies fail to capture the nuanced interactions between visual evidence and patient-specific clinical history that are central to real-world diagnostic reasoning. This work aims to address this limitation by enabling history-aware interpretation of chest X-rays, allowing patient context to directly influence visual attention and diagnostic outcomes.

To overcome the shortcomings of existing systems, we propose a history-conditioned multimodal diagnostic framework that integrates a lung-focused visual encoder with a ClinicalBERT-based clinical history encoder through token-level cross-modal attention and a learned consistency mechanism. Unlike conventional systems that treat imaging and clinical history as independent predictors, the proposed technique performs early fusion at the representation level, allowing clinical history to dynamically refine visual feature interpretation and suppress clinically inconsistent predictions.

The main contributions of this paper are summarized as follows:

- We propose a novel history-conditioned multimodal architecture that jointly models chest X-ray images and free-text clinical history for lung disease diagnosis.
- We introduce a token-level cross-modal attention mechanism that enables clinical history to guide visual attention within lung regions.
- We replace decision-level agreement with a learned image–history consistency strategy, improving patient-specific diagnostic reasoning.
- We provide an interpretable diagnostic framework suitable for clinical decision support without requiring paired radiology reports or structured clinical variables.

II. Related Work

Early studies establish that incorporating non-visual patient information alongside chest X-rays improves diagnostic reliability. Hsieh et al. demonstrate that feature-level fusion of chest X-rays and structured clinical parameters improves abnormality detection; however, their reliance on numeric clinical data and lack of cross-modal attention limit their ability to model rich clinical narratives (Hsieh et al., 2023). Extending this line of work, Khader et al. show that transformer-based multimodal fusion of chest radiographs and clinical parameters outperforms CNN-based approaches, yet their dependence on manually engineered structured features restricts applicability to real-world free-text medical histories (Khader et al., 2023).

Moving beyond structured data, several works explore vision–language alignment for medical understanding. Huang et al. introduce GLoRIA, a global–local vision–language alignment framework that improves label-efficient chest X-ray classification by grounding report sentences in image regions; however, the approach relies on paired radiology reports and does not incorporate patient-specific clinical history (Huang et al., 2021). Jahanian et al. further demonstrate that

joint vision–language transformers outperform decision-level multimodal fusion by enforcing semantic alignment between chest X-rays and radiology reports, suggesting that shared representations are more effective than independent predictors, though their reliance on paired image–report data limits general applicability (Jahanian et al., 2025).

To address data scarcity and pairing constraints, contrastive pretraining approaches have been proposed. Wang et al. introduce MedCLIP, a contrastive learning framework that leverages unpaired medical images and text while mitigating false negatives through knowledge-driven semantic matching, enabling data-efficient zero-shot and transfer performance; nevertheless, the approach focuses on general medical semantics rather than patient-specific reasoning (Wang et al., 2022). In parallel, Tiu et al. demonstrate that large-scale self-supervised learning enables expert-level chest X-ray pathology detection without annotations, but the absence of clinical context limits diagnostic specificity, highlighting the need for multimodal reasoning beyond visual cues alone (Tiu et al., 2022).

Several studies emphasize the importance of explicit medical knowledge and contextual reasoning in vision–language models. Wu et al. propose MedKLIP, a knowledge-enhanced vision–language pretraining framework that aligns chest X-ray images with structured disease concepts, improving label efficiency and semantic grounding; however, the approach lacks patient-specific clinical history integration (Wu et al., 2023). Zhang et al. similarly introduce a knowledge-enhanced visual–language framework that aligns chest X-ray regions with radiology findings and diseases, improving spatial grounding and classification robustness, yet still omits individualized patient context (Zhang et al., 2023).

Focusing specifically on diagnostic fusion, Yang et al. demonstrate that integrating free-text clinical history with chest X-ray images using cross-modal attention improves multi-label diagnosis by refining visual focus based on patient context; however, their framework assumes the availability of history text and remains limited to static, single-time-point inference (L. Yang et al., 2025). In contrast, Yang et al. address multimodal reasoning from a report-generation perspective, proposing a learned medical knowledge base that mediates image–language alignment to reduce hallucinations in radiology reports, though their approach prioritizes language generation over explicit diagnostic decision-making (S. Yang et al., 2023).

Beyond classification, multimodal reasoning has also been explored through language generation tasks. Aksoy et al. demonstrate that integrating clinical context with chest X-ray images improves semantic grounding and factual consistency in report generation; however, their focus on language output rather than diagnostic prediction limits direct applicability to decision support (Aksoy et al., 2024). Varol Arısoy et al. propose a vision-attention-driven framework that improves report coherence using cross-attention, memory mechanisms, and tag-guided generation, but ClinicalBERT is primarily used for tag extraction rather than modeling longitudinal patient history (Varol Arısoy et al., 2025). Nakasan shows that Vision Transformers combined with biomedical language models improve chest X-ray captioning quality, yet the unidirectional vision-to-language design and lack of patient context restrict its relevance for diagnostic reasoning (Patcharapanyawat et al., 2025).

Recent studies highlight additional dimensions often overlooked in static multimodal models. Bannur et al. demonstrate that incorporating temporal structure into biomedical vision–language models significantly improves performance, revealing a key limitation of single-time-point diagnostic systems that ignore disease progression (Bannur et al., 2023). Finally, Tanno et al. emphasize the importance of human–AI collaboration in radiology, arguing that vision–language models should function as decision-support tools rather than autonomous agents, underscoring the need for interpretability, controllability, and clinically aligned multimodal systems (Tanno et al., 2025).

Despite significant progress in chest X-ray–based diagnosis, existing multimodal systems exhibit several critical limitations that hinder their applicability to real-world clinical decision support. Image-only models, even when strengthened through large-scale self-supervised learning, remain context-blind and fail to resolve clinically ambiguous visual patterns without patient history. Structured-data fusion approaches improve performance but rely on manually engineered numeric features that cannot capture the richness of free-text clinical narratives. Vision–language alignment and knowledge-enhanced models, while effective in semantic grounding, often depend on paired radiology reports or static disease concepts and thus lack patient-specific reasoning. Moreover, many multimodal diagnostic pipelines employ late decision-level fusion, where image and text predictions are combined post hoc, preventing clinical history from directly influencing visual interpretation and often resulting in visually plausible but clinically inconsistent predictions.

To address these gaps, we propose a history-conditioned multimodal diagnostic framework that replaces decision-level agreement with early, token-level cross-modal fusion, allowing free-text clinical history encoded via ClinicalBERT to dynamically guide attention over lung-region visual tokens. In addition, a learned image–history consistency mechanism is introduced to model agreement between modalities in a soft, data-driven manner rather than enforcing rigid constraints. This design enables patient-specific, interpretable, and robust diagnosis without requiring paired radiology reports or structured clinical variables, directly addressing the shortcomings identified in prior approaches.

III. Proposed Framework

In this section, we present our history-conditioned multimodal diagnostic system whose objective is to predict multi-label thoracic diseases from chest X-rays while using free-text clinical history to refine visual attention and suppress clinically inconsistent predictions, outperforming late-fusion/decision-agreement baselines.

A. Data Input and Cleaning

Let $D = \{(x_i, t_i, y_i)\}_{i=1}^{N}$ denote the multimodal dataset of size N, where each sample consists of:

- an X-ray image x_i
- a corresponding clinical history text t_i
- a multi-label target vector $y \epsilon \{0,1\}^{14}$

Prior to model training, a data preprocessing stage is applied to ensure consistency and reliability of the inputs. This stage includes:

- removal of invalid or corrupted X-ray images
- normalization of label annotations to a fixed 14-class format
- handling of missing clinical history by assigning a special *NO_HISTORY* token or an empty-text placeholder

These preprocessing steps collectively produce a clean and standardized dataset suitable for subsequent multimodal feature extraction and fusion.

B. Image Preprocessing and Lung-ROI Extraction

To encourage the model to focus on anatomically relevant regions and reduce shortcut learning from image borders, textual markers, and acquisition artifacts, an explicit lung ROI extraction stage is applied to each chest X-ray image.

Let $x_i \in \mathbb{R}^{H \times W}$ denote the input chest radiograph corresponding to sample i. The objective of this module is to isolate the lung fields and suppress irrelevant background information prior to visual feature encoding.

A lung localization function is defined as:

$$f_{\text{ROI}}(\cdot) : \text{Image} \rightarrow \text{ROI}. \tag{1}$$

Two alternative strategies are considered for ROI extraction.

(1) Segmentation-based approach. A pretrained lung segmentation model (e.g., U-Net) is used to generate a binary lung mask:

$$m_i = f_{\text{seg}}(x_i), \tag{2}$$

where $m_i \in \{0,1\}^{H \times W}$ denotes lung pixels. Regions outside the mask are removed or suppressed, and the remaining lung area is cropped to form the lung-focused image.

(2) Bounding-box detection approach. Alternatively, a lightweight object detector (e.g., YOLO or RetinaNet) predicts bounding boxes

$$b_i = (x_{\min}, y_{\min}, x_{\max}, y_{\max}), \tag{3}$$

corresponding to the left and right lung regions. The predicted bounding boxes are used to crop the lung ROI from the original image.

The output of this module is the lung-focused image

$$x_i^{\text{lung}} \in \mathbb{R}^{H' \times W'}, \tag{4}$$

and optionally its associated lung mask m_i, which are subsequently passed to the visual encoder.

C. Visual Encoder

Given the lung-focused image x_i^{lung} obtained from the ROI extraction stage, a visual encoder is employed to convert the image into a sequence of discriminative visual tokens.

Let $x_i^{\text{lung}} \in \mathbb{R}^{H' \times W'}$ denote the lung ROI image. The objective of this module is to transform spatial visual information into a tokenized representation suitable for multimodal fusion.

A visual encoding function is defined as:

$$f_{\text{vis}}(\cdot) : \text{Image} \rightarrow \mathbb{R}^{N_v \times d_v}. \tag{5}$$

Two alternative visual encoding strategies are considered.

(1) Vision Transformer (ViT)-based encoding. The lung ROI image is divided into fixed-size non-overlapping patches. Each patch is linearly projected into a d_v-dimensional embedding, and positional encodings are added to preserve spatial structure. This results in a sequence of patch tokens:

$$V_i = \{v_{i1}, v_{i2}, \ldots, v_{iN_v}\}, \quad v_{ij} \in \mathbb{R}^{d_v}. \tag{6}$$

(2) CNN-based encoding. Alternatively, a convolutional backbone extracts a high-level feature map from the lung ROI. The spatial feature map is flattened across spatial dimensions, where each spatial location corresponds to a region token.

The final output of the visual encoder is a matrix of visual tokens:

$$V_i \in \mathbb{R}^{N_v \times d_v}, \tag{7}$$

where N_v denotes the number of visual tokens and d_v represents the embedding dimension. These tokens provide a compact yet expressive representation of lung structures and pathological cues, and serve as visual inputs for subsequent multimodal fusion and cross-attention mechanisms.

D. Clinical History Encoder (ClinicalBERT)

Given the clinical history text t_i associated with each patient sample, the textual information is encoded using a domain-specific language model designed for medical narratives.

Let $t_i = \{w_1, w_2, \ldots, w_{N_t}\}$ denote the tokenized clinical history sequence. The objective of this module is to transform free-text medical histories into dense contextual embeddings that capture semantic and clinical dependencies.

Clinical Text Encoding. We employ a pretrained ClinicalBERT (or BioClinicalBERT) model, denoted by the function

$$f_{\text{text}}(\cdot) : \text{Text} \rightarrow \mathbb{R}^{N_t \times d_t},$$

which processes the input token sequence and produces contextualized representations from the final hidden layer of the transformer.

Each token is mapped to a d_t-dimensional embedding, resulting in

$$T_i = \{t_{i1}, t_{i2}, \ldots, t_{iN_t}\}, \quad t_{ij} \in \mathbb{R}^{d_t}.$$

In addition to token-level embeddings, the special classification token `[CLS]` is optionally retained as a global summary representation:

$$t_i^{\text{cls}} \in \mathbb{R}^{d_t},$$

which captures an overall representation of the patient's clinical history.

The final output of this module consists of the text token matrix

$$T_i \in \mathbb{R}^{N_t \times d_t},$$

E. Cross-Modal Fusion

To integrate visual and textual information, we perform token-level cross-modal fusion using a cross-attention mechanism. The objective of this module is to allow clinical history to guide the model's attention toward diagnostically relevant regions within the lung images. Let

$$V_i \in \mathbb{R}^{N_v \times d_v} \quad \text{and} \quad T_i \in \mathbb{R}^{N_t \times d_t} \tag{8}$$

denote the visual and textual token sequences, respectively. Prior to fusion, both modalities are projected into a shared embedding space of dimension d using linear transformation layers.

Cross-Modal Attention Fusion. We define a cross-attention function

$$f_{\text{cross}}(\cdot) : (T_i, V_i) \to (T_i^{\text{fused}}, V_i^{\text{fused}}) \tag{9}$$

which enables information exchange between the two modalities at the token level.

In the primary fusion direction, clinical history tokens serve as queries, while visual tokens act as keys and values:

$$\text{Attention}(Q = T_i,\ K = V_i,\ V = V_i) \tag{10}$$

allowing textual context to modulate attention over lung image regions.

Optionally, a second symmetric cross-attention block is applied in the reverse direction:

$$\text{Attention}(Q = V_i,\ K = T_i,\ V = T_i) \tag{11}$$

enabling bidirectional interaction between image and text representations.

The outputs of the fusion module are the updated token sequences

$$T_i^{\text{fused}} \in \mathbb{R}^{N_t \times d} \quad \text{and} \quad V_i^{\text{fused}} \in \mathbb{R}^{N_v \times d} \tag{12}$$

which jointly encode visual evidence and clinical context. These fused representations are subsequently used for downstream diagnosis and prediction tasks.

F. Consistency Learning Module

To model agreement between visual evidence and clinical history in a flexible manner, we introduce a consistency learning module that replaces hard decision constraints with a learned compatibility mechanism.

Let $V_i^{\text{fused}} \in \mathbb{R}^{N_v \times d}$ and $T_i^{\text{fused}} \in \mathbb{R}^{N_t \times d}$ denote the fused visual and textual token representations obtained from the cross-modal attention module. The objective of this stage is to quantify the semantic consistency between the two modalities and regulate their joint contribution to diagnosis.

Compatibility Estimation. Global representations are first obtained via pooling operations over the fused token sequences. Let $\text{pool}(\cdot)$ denote a global pooling function (e.g., mean or attention pooling). The pooled features are concatenated with the global clinical summary token t_i^{cls} to form a joint representation:

$$z_i = [\,\text{pool}(V_i^{\text{fused}})\ ;\ \text{pool}(T_i^{\text{fused}})\ ;\ t_i^{\text{cls}}\,]. \tag{13}$$

A compatibility score is then computed using a multilayer perceptron (MLP) followed by a sigmoid activation:

$$s_i = \sigma\left(\text{MLP}(z_i)\right), \tag{14}$$

where $s_i \in [0, 1]$ represents the learned image–history compatibility.

Gated Prediction. The compatibility score is used to adaptively regulate the influence of multimodal information during classification. One simple gating formulation is given by:

$$\text{logits}_i = \text{logits}_{\text{img}} + s_i \cdot \text{logits}_{\text{fused}}, \tag{15}$$

where $\text{logits}_{\text{img}}$ denotes predictions from the image-only branch and $\text{logits}_{\text{fused}}$ corresponds to predictions from the fused representation.

Alternatively, the fused logits may be used directly while incorporating an auxiliary loss term that encourages higher compatibility between consistent visual and textual cues.

The outputs of this module include the scalar compatibility score s_i, which quantifies agreement between image and clinical history, and a gated fused representation used for final multi-label disease classification.

G. Multi-Label Classification Head

The final diagnosis stage performs multi-label disease prediction based on the fused multimodal representation obtained from the preceding modules.

Let $h_i^{\text{fused}} \in \mathbb{R}^d$ denote the global fused feature vector, obtained by applying a pooling operation over the gated fused token representations. This vector encodes integrated visual and clinical information relevant for disease prediction.

Multi-label Classification. A linear or multilayer perceptron (MLP) classification head is applied to the fused global representation to generate disease-wise prediction scores:

$$\mathbf{z}_i = W h_i^{\text{fused}} + b, \tag{16}$$

where $W \in \mathbb{R}^{14 \times d}$ and $b \in \mathbb{R}^{14}$ denote the learnable parameters of the classification head.

The final disease probabilities are obtained using a sigmoid activation function applied independently to each label:

$$\mathbf{p}_i = \sigma(\mathbf{z}_i), \tag{17}$$

where $\mathbf{p}_i \in [0, 1]^{14}$ represents the predicted probabilities for the 14 thoracic disease categories.

This formulation enables independent modeling of co-occurring pathologies, making it suitable for multi-label chest X-ray diagnosis tasks.

H. Training Objective

The model is trained in an end-to-end manner using a composite objective function that jointly optimizes diagnostic accuracy and cross-modal consistency.

Main Classification Loss. Given the predicted logits $\mathbf{z}_i$ and the ground-truth multi-label target vector $\mathbf{y}_i \in \{0,1\}^{14}$, the primary training objective is defined using a multi-label binary cross-entropy loss:

$$\mathcal{L}_{\text{cls}} = \text{BCEWithLogits}(\mathbf{z}_i, \mathbf{y}_i), \tag{18}$$

which independently penalizes incorrect predictions for each disease category.

Consistency Regularization. To encourage agreement between image-based and multimodal predictions, we introduce a compatibility-aware regularization term. Let $s_i \in [0,1]$ denote the learned image–history compatibility score, and let $\mathbf{z}_i^{\text{img}}$ represent logits obtained from the image-only branch.

The consistency loss is defined as:

$$\mathcal{L}_{\text{cons}} = (1 - s_i) \cdot \text{mean}\left(\left|\sigma(\mathbf{z}_i) - \sigma(\mathbf{z}_i^{\text{img}})\right|\right), \tag{19}$$

which penalizes disagreement between predictions when compatibility is low or high, depending on the chosen formulation.

Total Training Objective. The final optimization objective combines the classification and consistency terms:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda\, \mathcal{L}_{\text{cons}}, \tag{20}$$

where λ is a hyperparameter that controls the contribution of the consistency regularization term.

This joint objective enables the model to learn accurate multi-label predictions while maintaining coherent reasoning between visual evidence and clinical history.

I. Inference and Explainability

During inference, the proposed framework produces both diagnostic predictions and interpretable explanations that support clinical decision-making.

Inference Output. Given a chest X-ray image and its associated clinical history, the trained model generates disease probability estimates for all thoracic conditions, along with auxiliary interpretability signals.

The final predicted disease probabilities are given by:

$$\mathbf{p}_i \in [0,1]^{14}, \tag{21}$$

corresponding to the presence likelihood of each disease category.

In addition to diagnostic predictions, the model outputs the learned image–history compatibility score:

$$s_i \in [0,1], \tag{22}$$

which reflects the degree of agreement between visual evidence and clinical context.

Explainability Mechanisms. To enhance transparency, attention-based interpretability is extracted from the multimodal fusion layers. Visual explanations are obtained by projecting cross-attention weights onto lung patch tokens, producing spatial attention maps that highlight diagnostically relevant lung regions.

Similarly, textual explanations are derived from attention distributions over clinical history tokens, enabling identification of the most influential words or phrases contributing to the final diagnosis.

The inference stage therefore returns:

- multi-label disease probability predictions for 14 thoracic pathologies,
- the image–history compatibility score s_i,
- visual attention maps over lung regions,
- and top contributing clinical history tokens.

These outputs collectively provide both accurate predictions and interpretable evidence, facilitating trust, transparency, and clinical usability of the proposed multimodal diagnosis system.

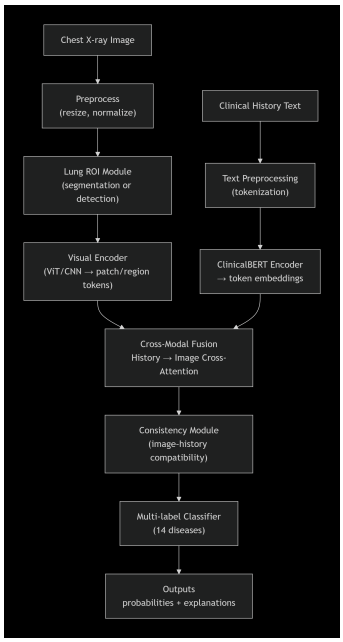

Fig. 1: Multimodal Chest X-ray Diagnosis Framework

Algorithm 1 Proposed Multimodal Chest X-ray Diagnosis Framework

Require: Chest X-ray image x, clinical history text t
Ensure: Disease probabilities $\mathbf{p} \in [0,1]^{14}$, compatibility score $s \in [0,1]$

/* 1) Data Cleaning & Standardization */
$x \leftarrow \textsc{ValidateImage}(x)$ {discard/skip if corrupted}
$\mathbf{y} \leftarrow \textsc{NormalizeLabels}(\mathbf{y})$ {14-label format (train-time)}
if t is missing **then**
  $t \leftarrow$ `NO_HISTORY`
end if
/* 2) Lung ROI Extraction */
if Segmentation-based ROI is used **then**
  $m \leftarrow f_{\text{seg}}(x)$ {lung mask}
  $x^{\text{lung}} \leftarrow \textsc{MaskOrCrop}(x, m)$
else
  $b \leftarrow f_{\text{det}}(x)$ {lung bounding box(es)}
  $x^{\text{lung}} \leftarrow \textsc{Crop}(x, b)$
end if
/* 3) Visual Encoding */
if ViT backbone is used **then**
  $V \leftarrow \textsc{PatchEmbed}(x^{\text{lung}})$ $\{V \in \mathbb{R}^{N_v \times d_v}\}$
else
  $F \leftarrow \textsc{CNNBackbone}(x^{\text{lung}})$
  $V \leftarrow \textsc{FlattenToTokens}(F)$ $\{V \in \mathbb{R}^{N_v \times d_v}\}$
end if
/* 4) Clinical History Encoding */
$\{T, t^{\text{cls}}\} \leftarrow \textsc{ClinicalBERT}(t)$ $\{T \in \mathbb{R}^{N_t \times d_t}\}$
/* 5) Cross-Modal Fusion (Token-level Cross-Attention) */
$\tilde{V} \leftarrow \textsc{Project}(V),\ \tilde{T} \leftarrow \textsc{Project}(T)$ {to common dim d}
$T^{\text{fused}} \leftarrow \textsc{CrossAttn}(Q = \tilde{T}, K = \tilde{V}, V = \tilde{V})$
$V^{\text{fused}} \leftarrow \textsc{CrossAttn}(Q = \tilde{V}, K = \tilde{T}, V = \tilde{T})$ {optional symmetric block}
/* 6) Consistency Learning (Compatibility + Gating) */
$v_g \leftarrow \textsc{Pool}(V^{\text{fused}}),\ t_g \leftarrow \textsc{Pool}(T^{\text{fused}})$
$z \leftarrow [v_g\ ;\ t_g\ ;\ t^{\text{cls}}]$
$s \leftarrow \sigma(\text{MLP}(z))$ {image–history compatibility}
/* 7) Multi-label Prediction */
$\mathbf{z}_{\text{img}} \leftarrow \textsc{HeadImg}(\textsc{Pool}(V))$ {optional image-only logits}
$\mathbf{z}_{\text{fused}} \leftarrow \textsc{HeadFused}(h^{\text{fused}})$ $\{h^{\text{fused}} = \textsc{Pool}(\textsc{Gate}(V^{\text{fused}}, T^{\text{fused}}, s))\}$
$\mathbf{z} \leftarrow \mathbf{z}_{\text{img}} + s \cdot \mathbf{z}_{\text{fused}}$ {one simple gating option}
$\mathbf{p} \leftarrow \sigma(\mathbf{z})$
return $\mathbf{p}, s$

References

Aksoy, N., Sharoff, S., Baser, S., Ravikumar, N., & Frangi, A. F. (2024). Beyond images: An integrative multimodal approach to chest x-ray report generation. *Frontiers in radiology*, *4*, 1339612.

Bannur, S., Hyland, S., Liu, Q., Perez-Garcia, F., Ilse, M., Castro, D. C., Boecking, B., Sharma, H., Bouzid, K., Thieme, A., et al. (2023). Learning to exploit temporal structure for biomedical vision-language processing. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 15016–15027.

Hsieh, C., Nobre, I. B., Sousa, S. C., Ouyang, C., Brereton, M., Nascimento, J. C., Jorge, J., & Moreira, C. (2023). Mdf-net for abnormality detection by fusing x-rays with clinical data. *Scientific Reports*, *13*(1), 15873.

Huang, S.-C., Shen, L., Lungren, M. P., & Yeung, S. (2021). Gloria: A multimodal global-local representation learning framework for label-efficient medical image recognition. *Proceedings of the IEEE/CVF international conference on computer vision*, 3942–3951.

Jahanian, M., Karimi, A., Eraghi, N. O., & Zarafshan, F. (2025). Multimodal transformers for joint diagnosis and retrieval from chest x-rays and radiology reports. *Informatics in Medicine Unlocked*, 101672.

Khader, F., Müller-Franzes, G., Wang, T., Han, T., Tayebi Arasteh, S., Haarburger, C., Stegmaier, J., Bressem, K., Kuhl, C., Nebelung, S., et al. (2023). Multimodal deep learning for integrating chest radiographs and clinical parameters: A case for transformers. *Radiology*, *309*(1), e230806.

Patcharapanyawat, S., Nakasan, C., & Chantrapornchai, C. (2025). Chest x-ray image captioning using vision transformer and biomedical language models with gru and optuna tuning. *Science & Technology Asia*, 154–162.

Tanno, R., Barrett, D. G., Sellergren, A., Ghaisas, S., Dathathri, S., See, A., Welbl, J., Lau, C., Tu, T., Azizi, S., et al. (2025). Collaboration between clinicians and vision–language models in radiology report generation. *Nature Medicine*, *31*(2), 599–608.

Tiu, E., Talius, E., Patel, P., Langlotz, C. P., Ng, A. Y., & Rajpurkar, P. (2022). Expert-level detection of pathologies from unannotated chest x-ray images via self-supervised learning. *Nature biomedical engineering*, *6*(12), 1399–1406.

Varol Arısoy, M., Arısoy, A., & Uysal, İ. (2025). A vision attention driven language framework for medical report generation. *Scientific Reports*, *15*(1), 10704.

Wang, Z., Wu, Z., Agarwal, D., & Sun, J. (2022). Medclip: Contrastive learning from unpaired medical images and text. *Proceedings of the Conference on Empirical Methods in Natural Language Processing. Conference on Empirical Methods in Natural Language Processing*, *2022*, 3876.

Wu, C., Zhang, X., Zhang, Y., Wang, Y., & Xie, W. (2023). Medklip: Medical knowledge enhanced language-image pre-training for x-ray diagnosis. *Proceedings of the IEEE/CVF international conference on computer vision*, 21372–21383.

Yang, L., Wan, Y., & Pan, F. (2025). Enhancing chest x-ray diagnosis with a multimodal deep learning network by integrating clinical history to refine attention. *Journal of Imaging Informatics in Medicine*, 1–16.

Yang, S., Wu, X., Ge, S., Zheng, Z., Zhou, S. K., & Xiao, L. (2023). Radiology report generation with a learned knowledge base and multi-modal alignment. *Medical Image Analysis*, *86*, 102798.

Zhang, X., Wu, C., Zhang, Y., Xie, W., & Wang, Y. (2023). Knowledge-enhanced visual-language pre-training on chest radiology images. *Nature Communications*, *14*(1), 4542.